

# PHYO THANT KYAW

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Nationality: Myanmar | Visa: Non-Immigrant ED Plus | **Authorized: 16-week Co-op, No Sponsorship Required**

Eligible: 1-Year Post-Grad Extension + BOI IT Specialist Tier | Languages: English (Professional), Thai (Basic), Burmese (Native)

## SUMMARY

Data engineering specialist with expertise in designing automated ETL pipelines, complex feature engineering (time-series lag, topographical indicators), and Python-based data infrastructure on real-world environmental datasets.

**Seeking a 16-week Co-op (January 2027)** to architect scalable data infrastructure and optimize end-to-end data pipelines at a Tier-1 Thai tech organization.

## EDUCATION

**Mae Fah Luang University (MFU)** | B.Eng., Computer Engineering | GPAX: **3.84**

*Expected 2027*

Mandatory 16-week Cooperative Education: **January 2027**

**Mandalay Technological University (MTU)** | B.Eng., Electronics Engineering | GPA: 3.04 *Dec 2016 – Sep 2019 (Year 3)*

## TECHNICAL PROJECTS

**MFU PM2.5 GeoAI Warning System (Senior Project)** | [Live Demo](#) | [GitHub](#)

*Jan – Jul 2026*

- **Architected** a 4-model comparative forecasting pipeline (LightGBM, XGBoost, SVR, MLR) using a strict chronological train/validation/test split, with LightGBM emerging as champion at 85.90%  $R^2$  (MAE: 3.21) against a 5-year localized Chiang Rai dataset.
- **Engineered** distance-decay Fire Pressure features across 25km/50km/100km radiuses from NASA FIRMS (VIIRS) satellite hotspot data, integrated with OpenWeatherMap and Air4Thai meteorological APIs.
- **Implemented** data validation checkpoints and a graceful fallback architecture across the ingestion pipeline, ensuring stable data delivery during API downtime.
- **Deployed** a production Streamlit dashboard with GISTDA Sphere API spatial visualization and automated Gemini AI multilingual advisories.

**Chiang Mai Tri-Node Zone: Geo-Predictive AI & Mitigation Platform** | [Live Demo](#) | [GitHub](#)

*Mar – May 2026*

- **Engineered** Z-score spatial analytics and wind-direction correlation pipelines across 3 Chiang Mai monitoring nodes to detect micro-level air quality shifts in real time.
- **Transformed** raw multi-node NASA FIRMS (Fire Radiative Power) and OpenWeatherMap records using Pandas—standardizing coordinate systems, resolving missing values, and structuring outputs for LLM prompt injection.
- **Structured** and stored multi-node geospatial data for seamless querying and integration with Google Gemini LLM for real-time strategy synthesis.
- **Built** robust Python data delivery pipelines feeding an interactive geospatial threat-visualization dashboard with zero manual refresh.

## EXPERIENCE

**Work Assistant — Web & Data Projects** | MFU Learning Innovation Institute (MLii)

*May 2026 – Jul 2026*

- **Engineered** a secure, Thai PDPA-compliant PostgreSQL database architecture via Supabase to ingest, encrypt, and store clinical assessment data.
- **Architected** a backend data flow integrating a Zero-Budget Hybrid AI Engine (Rule-based + Gemini LLM), optimizing API response times with Upstash Redis caching.
- **Authored** the complete Software Requirements Specification (SRS), structuring RESTful API endpoints, robust data validation pipelines (Zod), and Role-Based Access Control (RBAC).

## TECHNICAL SKILLS

**Data Engineering:** ETL Pipeline Design, Data Preprocessing, Feature Engineering, API Integration (REST), Data Validation, Apache Spark, Hadoop

**Programming & DB:** Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL

**Cloud & Infrastructure:** Google Cloud Platform (GCP), Docker, Git, GitHub

**Analytics & Web:** LightGBM, XGBoost, Streamlit, LangChain, Google Gemini API

**Certifications:** IBM Data Engineering Professional | Google Data Analytics | Google Cloud: Data Lakes & Warehouses | UC Davis: SQL for Data Science | Google AI Essentials, Prompting Essentials